

AI LAB 99
WEEK 5 • DAY 3 • MODULE 10

CUSTOMER INTELLIGENCE PLATFORM

Professional Project Report

K-Means Customer Segmentation • Behavioral Profiling • Marketing Strategy

Project Title	Customer Intelligence Platform
Assigned By	AI LAB 99
Course Module	Week 5 – Day 3 – Module 10
Model Type	K-Means Unsupervised Clustering (Scikit-Learn)
Application Framework	Streamlit Interactive Dashboard
Dataset	Customer Personality Final Dataset (2,038 records × 40 features)
Report Date	August 2026

This report documents the complete end-to-end pipeline of the Customer Intelligence Platform, covering model loading, feature recovery, real-time segment prediction, customer persona generation, and actionable marketing strategy recommendations.

1. Executive Summary

The Customer Intelligence Platform is an end-to-end machine learning solution designed to segment customers, generate behavioral personas, and produce data-driven marketing recommendations. The system leverages a pre-trained K-Means clustering model to assign customers into meaningful groups based on demographic, transactional, and behavioral attributes.

Key achievements of this module include:

- Successful recovery and loading of the trained K-Means model (2 clusters, 40 input features).
- Automatic identification of exact model feature names from the saved model artifact.
- Real-time customer segment prediction using live input features.
- Generation of detailed customer personas including purchase behavior, product preference, campaign response, and risk level.
- Automated marketing strategy recommendations (channel, offer, timing, cross-sell, up-sell, retention).
- Deployment-ready Streamlit application with multi-page navigation for exploration, prediction, and download.

2. Project Objectives

1. **Model Recovery:** Locate, load, and validate the saved K-Means model and associated preprocessing artifacts.

2. **Feature Alignment:** Recover the exact 40 feature names expected by the model and align them with the final dataset.
3. **Segment Prediction:** Enable real-time prediction of customer segment for any new or existing customer profile.
4. **Persona Generation:** Build a rich customer persona covering income, spending, recency, activity, risk, and product preference.
5. **Marketing Intelligence:** Derive actionable recommendations for communication channel, offers, campaign timing, cross-selling, and retention.
6. **Interactive Delivery:** Package all insights into a professional Streamlit dashboard for business stakeholders.

3. Dataset Overview

The primary dataset used is the Customer Personality Final Dataset, prepared during earlier feature engineering modules. Key statistics:

Attribute	Value
Total Records (Customers)	2,038
Total Features	40
Number of Clusters (K-Means)	2
Data Source Path	Week 3 / Module 5 – Feature Engineering
File Name	Customer_Personality_Final_Dataset.csv

3.1 Feature Categories

- **Demographics:** Education, Marital_Status, Age, Customer_Age, Partner, Family_size, Total_Children
- **Financial:** Income, Total_spending, Average Spending per Purchase
- **Purchase Behavior:** NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth, Total_Purchase
- **Product Spending:** MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds
- **Campaign Response:** AcceptedCmp1–5, Response, Total_Campaign
- **Engagement & Risk:** Recency, Customer_Tenure, Digital_Engagement, Deal_Dependency, Preferred_Shopping_Channel, Product_Preference, Customer_Activity_Level, Complain

4. Model Details

The core predictive component is a pre-trained K-Means clustering model saved as kmeans_model.pkl. Model inspection confirmed the following parameters:

Parameter	Value
Algorithm	Lloyd (standard K-Means)
Number of Clusters (n_clusters)	2
Input Features (n_features_in_)	40
Cluster Centers Shape	(2, 40)
Initialization	k-means++
n_init	10
Max Iterations	300

Random State	42
Tolerance	0.0001

Feature names were successfully recovered directly from the model attribute `feature_names_in_`, ensuring exact alignment between training-time features and inference-time inputs.

5. Segment Analysis Results

After applying the K-Means model to the final dataset, customers were assigned to one of two segments. The primary analyzed segment (Predicted Cluster 0) exhibits the following profile:

5.1 Segment Summary (Cluster 0)

Metric	Value
Customers in Segment	1,111
Average Income (scaled)	0.67
Average Total Spending (scaled)	0.81
Average Recency (scaled)	0.01
Risk Level	Very Low
Dominant Activity Level	High (≈ 1.28)

5.2 Purchase Behavior

Average values (scaled) for key purchase channels within the segment:

Purchase Channel	Average Value
NumCatalogPurchases	0.731
NumStorePurchases	0.706
NumWebPurchases	0.657
NumDealsPurchases	0.040
NumWebVisitsMonth	-0.395

5.3 Product Preference

Preferred product categories ranked by average spending:

Product Category	Average Spending
MntWines (Primary)	0.760
MntMeatProducts (Secondary)	0.758
MntFruits	0.578
MntFishProducts	0.572
MntGoldProds	0.560
MntSweetProducts	0.553

6. Customer Persona

Based on the segment analysis, the following high-level persona was constructed for Cluster 0:

Persona Attribute	Description
Predicted Cluster	0
Segment Size	1,111 customers
Income Profile	Above-average (scaled mean 0.67)
Spending Profile	High total spending (scaled mean 0.81)
Recency	Very recent engagement (scaled 0.01)
Activity Level	Predominantly High
Churn Risk	Very Low
Preferred Product	Wines (MntWines)
Secondary Product	Meat Products

7. Marketing Strategy Recommendations

Rule-based logic was applied on top of the segment statistics to produce concrete marketing actions:

Recommendation	Details
Marketing Message	Enjoy Exclusive Rewards and Premium Products Selected for You
Personalized Offer	Exclusive Product Offer
Campaign Timing	Weekly
Campaign Frequency	Bi-Weekly
Primary Product	MntWines
Secondary Product	MntMeatProducts
Cross-Selling	Recommend MntMeatProducts with MntWines
Upselling	Offer premium or higher-value MntWines products
Discount Strategy	Low Discount / Value-Based Offer
Retention Strategy	Loyalty rewards and exclusive benefits
Risk Level	Very Low

8. Streamlit Application Architecture

A production-ready Streamlit application (app.py) was developed to deliver the intelligence platform to end users. The application automatically discovers required artifacts (Final Dataset.csv, kmeans_model.pkl, day3_results.pkl) under the project root and exposes the following pages:

- **Home:** KPI cards, system health, and file discovery status.
- **Dashboard:** Dataset preview, segment distribution, and numeric statistics.
- **Customer Segments:** Interactive exploration of any segment with key metrics.
- **Customer Search:** Full-text search across all customer attributes.
- **Customer Prediction:** Real-time segment inference using the loaded K-Means model with safe numeric inputs.
- **Visualizations:** Feature-level charts and descriptive statistics.
- **Marketing Strategy:** Display of Day-3 business recommendations loaded from day3_results.pkl.
- **Download:** Export of the final dataset and Day-3 results for offline use.

8.1 Technical Stack

- Python 3.11 + Pandas, NumPy, Joblib, Scikit-Learn
- Streamlit for interactive multi-page UI
- Cached data & model loading for performance
- Safe number-input helpers that respect actual data ranges (prevents min_value crashes)
- Automatic feature-name recovery from model.feature_names_in_

9. Artifacts & Deliverables

Artifact	Description
kmeans_model.pkl	Trained K-Means model (2 clusters, 40 features)
feature_list.pkl / standard_scaler.pkl	Supporting preprocessing artifacts (Module 6)
Customer_Personality_Final_Dataset.csv	Final engineered dataset (2,038 × 40)
day3_results.pkl	Serialized Day-3 persona, summary & marketing outputs
app.py	Streamlit Customer Intelligence Platform
proj.ipynb	Full analysis notebook documenting the workflow

10. Conclusion

Module 10 (Week 5 – Day 3) successfully closes the customer segmentation pipeline by moving from offline model development to an interactive, production-ready intelligence platform. The system demonstrates:

- Reliable recovery of a previously trained unsupervised model and its exact feature contract.
- Accurate real-time segment assignment for new customer profiles.
- Generation of business-ready personas and marketing playbooks.
- A polished Streamlit interface that can be handed directly to marketing and analytics stakeholders.

The platform is now ready for further extension (additional clusters, supervised propensity models, or A/B testing of recommended offers) while remaining fully reproducible from the saved artifacts.

— End of Report —

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